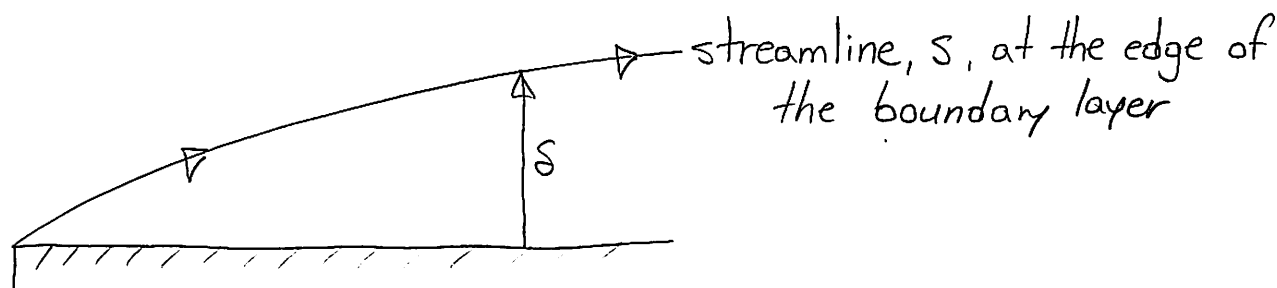


Now our life becomes much easier. We can use one more trick up our sleeves:



At the streamline, our x-momentum equation becomes:

$$u \frac{\partial u}{\partial x} + \underbrace{v \frac{\partial u}{\partial y}}_{0 \text{ at } s \text{ (no shear)}} = - \frac{1}{\rho} \frac{\partial p}{\partial x} + \underbrace{\nu \left(\frac{\partial^2 u}{\partial y^2} \right)}_{0 \text{ at } s \text{ (no shear)}}$$

At s , $u = V_\infty$

$$V_\infty \frac{\partial V_\infty}{\partial x} + \frac{1}{\rho} \frac{\partial p}{\partial x} = 0 \xrightarrow{\text{integrate}} \boxed{\rho + \frac{1}{2} \rho V_\infty^2 = \text{constant}}$$

↳ Bernoulli equation

And since we know $\rho \neq f(y)$, this is also true inside the boundary layer.

We also know that: $V_\infty = \text{constant}$, so $\frac{\partial V_\infty}{\partial x} = 0$

$$\therefore \frac{1}{\rho} \frac{\partial p}{\partial x} = 0 \Rightarrow \boxed{p(x) = \text{constant}} \text{ or } \boxed{\frac{\partial p}{\partial x} = \frac{\partial \bar{p}}{\partial x} = 0}$$

Note, this is only true for a flat plate. Not true for cylinders or spheres where $V_\infty \neq \text{constant}$.

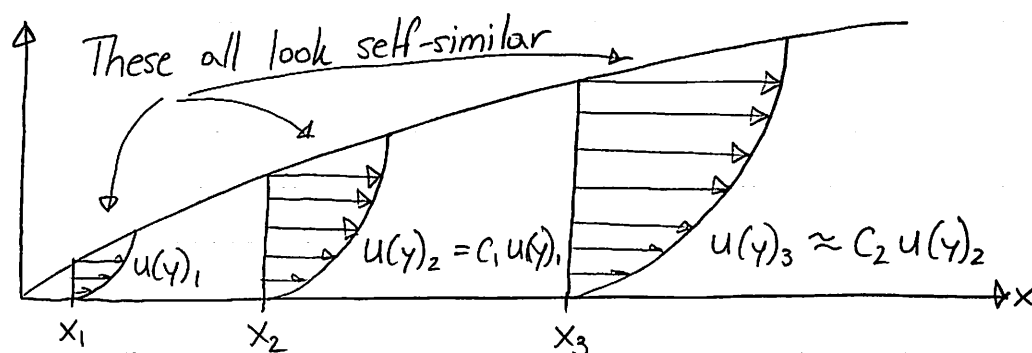
Now we can recast our x-momentum equation:

$$\boxed{u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = \nu \frac{\partial^2 u}{\partial y^2}}$$

Our boundary conditions are: 1) $u=v=0$ at $y=0$
 2) $u=V_\infty$ at $y \rightarrow \infty$

Just like in transient conduction, we want to convert our simpler but still complex PDE into an ODE.

Cannot use separation of variables (infinite domain, $y \rightarrow \infty$)
 Let's try the similarity solution:



$\downarrow$ A function ϕ that is a function of η only.
 $\frac{u}{V_\infty} = \phi(\eta)$; where $\eta = \frac{y}{\delta}$
 $\downarrow$ Scales from 0 to 1 $\downarrow$ Scales from 0 to 1

But how do we define δ ?

Let's look back at our dimensionless momentum equation:

$$\underbrace{\bar{u} \frac{\partial \bar{u}}{\partial \bar{x}} + \bar{v} \frac{\partial \bar{u}}{\partial \bar{y}}}_{\text{We know these are both on the order 1.}} = \underbrace{\frac{1}{Re_L} \frac{\partial^2 \bar{u}}{\partial \bar{y}^2}}_{\text{This must also be on the order 1.}}$$

We know these are both on the order 1. Also: $\bar{u} \sim V_\infty$

$$\bar{y} = \frac{y}{L} \sim \frac{\delta}{L} \Rightarrow \frac{\partial \bar{u}}{\partial \bar{y}} \sim \frac{\delta}{L}$$

$$\frac{U}{V_\infty L} \cdot V_\infty \cdot \left(\frac{L}{\delta}\right)^2 \sim 1 \Rightarrow \frac{\delta}{L} \sim \frac{1}{\sqrt{Re_L}} \text{ or } \frac{\delta}{x} \sim \frac{1}{\sqrt{Re_x}}$$

So we end up with: $\delta \sim \sqrt{\frac{UX'}{V_\infty}}$ or $\eta = y \sqrt{\frac{V_\infty}{UX}}$

↳ We'll see a much easier way to get this later.

So what do we do about V ?
Using continuity:

$$\frac{\partial \bar{u}}{\partial x} + \frac{\partial \bar{v}}{\partial y} = 0 \Rightarrow \bar{u} = \phi(\eta) = \frac{U}{V_\infty}$$

$$\frac{\partial \bar{u}}{\partial x} = \frac{\partial \bar{u}}{\partial \eta} \cdot \frac{\partial \eta}{\partial x}$$

$$\frac{\partial \bar{u}}{\partial x} = -\frac{1}{2x} \eta \phi' \Rightarrow \text{Back substitute into continuity}$$

$$\frac{\partial \bar{v}}{\partial y} = \frac{1}{2x} \eta \phi'$$

$$\bar{v} = \frac{1}{2x} \int_0^\eta \phi' \eta dy = \frac{1}{2x} \underbrace{\sqrt{\frac{xU}{V_\infty}}}_{\frac{1}{2\sqrt{Re_x}}} \int_0^\eta \phi' \eta d\eta$$

$d\eta = dy \sqrt{\frac{V_\infty}{UX}}$

We need to solve this integral:

$$\begin{aligned} \int_0^\eta \eta \phi' d\eta &= u v - \int v du \quad (\text{Integration by parts}) \\ &= \eta \phi - \int_0^\eta \phi d\eta \end{aligned}$$

$\int u dv = uv - \int v du$
 $u = \eta, v = \phi$

let $\int_0^\eta \phi d\eta = F(\eta)$

Now we have: $\bar{v} = \frac{1}{2\sqrt{Re_x}} (\eta F' - F)$

Back substituting into our initial PDE

$$\bar{u} \frac{\partial \bar{u}}{\partial x} + \bar{v} \frac{\partial \bar{u}}{\partial y} = \frac{1}{Re_L} \frac{\partial^2 \bar{u}}{\partial y^2}$$

$\uparrow$ $\nwarrow$
 $\bar{u} = F'$ $\frac{\partial \bar{u}}{\partial x} = \frac{\partial \bar{u}}{\partial \eta} \cdot \frac{\partial \eta}{\partial x} = F'' \cdot \frac{\partial \eta}{\partial x} \dots$ (Back substitute the rest)

Simplifying, we will obtain:

$$\boxed{F''' + \frac{1}{2} F F'' = 0} \quad \text{where } F = \int_0^\eta \phi d\eta, \quad \eta = y \sqrt{\frac{V_\infty}{xL}}$$

$\hookrightarrow$ ODE!

Our boundary conditions now become:

- 1) At the wall ($y=0, \eta=0$), $F' = F = 0$
- 2) Outside the boundary layer ($y \rightarrow \infty, \eta \rightarrow \infty$), $F' = 1$

Rewriting our equation in terms of $\bar{u}$, F , and η :

$$\boxed{\frac{\partial^2 \bar{u}}{\partial \eta^2} + \frac{1}{2} F \frac{\partial \bar{u}}{\partial \eta} = 0} \quad (1)$$

To solve this ODE, we can assume an infinite series sol.:

$$\begin{aligned} F &= a_0 + a_1 \eta + a_2 \eta^2 + a_3 \eta^3 + \dots \\ F' &= a_1 + 2a_2 \eta + 3a_3 \eta^2 + \dots \\ F'' &= 2a_2 + 6a_3 \eta + \dots \\ F''' &= 6a_3 + \dots \end{aligned}$$

Back substitute into (1) and solve for the coefficients

$$\underbrace{(\dots)}_0 \eta^0 + \underbrace{(\dots)}_0 \eta' + \underbrace{(\dots)}_0 \eta^2 + \underbrace{(\dots)}_0 \eta^3 + \dots + \underbrace{(\dots)}_0 \eta^n = 0$$

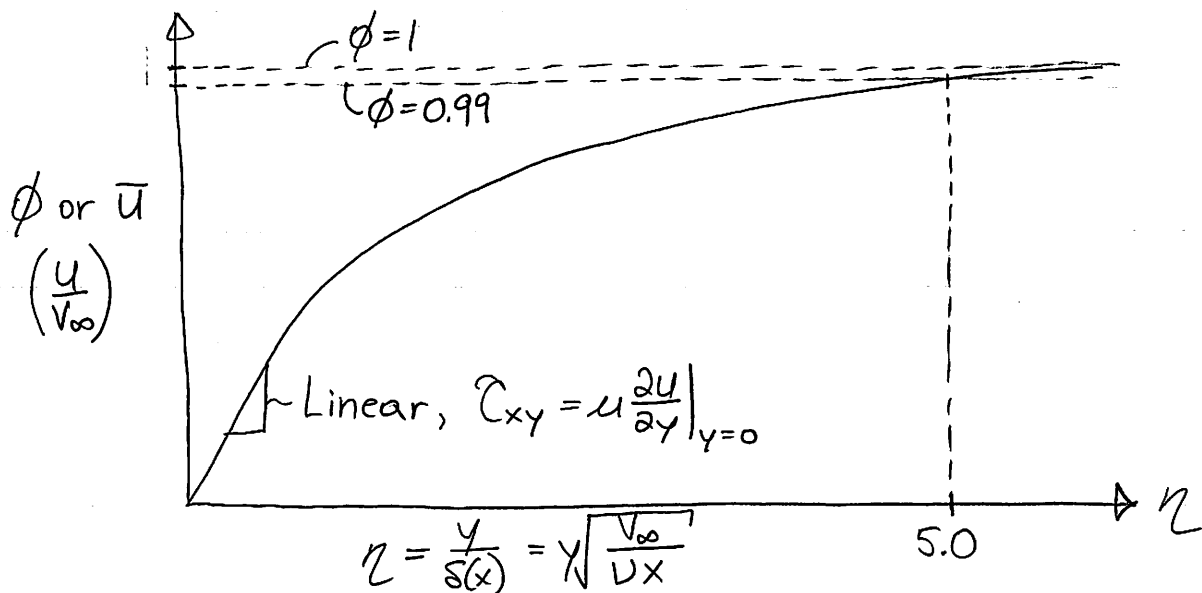
You obtain a recursion formula that relates your constants:

$$F = \frac{a_2 \eta^2}{2!} - \frac{a_2 \eta^5}{2 \cdot 5!} + \frac{11}{4} \frac{a_2^3 \eta^8}{8!} + \dots$$

$a_2 = 0.332 \Rightarrow$ Heinrich Blasius solved this in 1911 for his PhD work with Ludwig Prandtl.

$$\text{So } F = \int_0^\eta \phi d\eta, \quad F' = \phi = \bar{u} = \frac{u}{V_\infty}$$

We can now plot our result:



From our solution, we can solve for the hydrodynamic b.l. thick.

$$5.0 = \delta \sqrt{\frac{V_\infty}{\nu x}} = \frac{\delta}{x} \sqrt{Re_x} \Rightarrow \boxed{\delta(x) = \frac{5x}{\sqrt{Re_x}}}$$

$\hookrightarrow$ Boundary layer thickness